

Technical Appendix

The Missing Band Is a Distribution: Do Generative Bandwidth-Extension Models Sample It?

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This appendix accompanies the two-page paper. It gives the protocol, the model and training, exact metric definitions, the derivations behind the argument, the audit of the released models, every metric for every model on every test set, and the limitations. All numbers are generated from `results/results.json` by `results/make_tables.py`; none are typed by hand. The code that produced them is in `code/`, and listening examples are in `audio/`.

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1 What each claim rests on

Claim in the paper	Evidence	Where
The scoring harness reproduces the released models’ published numbers	LSD/VisQOL of FLowHigh and AP-BWE within 0.011 and 0.02 of their papers	Tab. 4
With a correct low band the squared error is $E(1+p-2c)$; an empty band maximises SNR when $c \approx 0$	Derivation; measured ceiling; NU-Wave 2’s published SNR bounds its level	§5.1, Tab. 11, Fig. 2
No model recovers the high-band waveform	Coherent fraction $c < 0.02$ for every model, 240 utterances	Tab. 12
When $c \approx 0$, the SNR gap of independent draws is a function of level	Derivation; measured gap against the independent value	§5.2, Tab. 14
FLowHigh is a regressor: one Euler step returns the conditional mean	Code path, derivation, gap 0.08 against 1.61, rank tails 41% / 46%	§5.5, §6.1, Fig. 3
NU-Wave 2 samples but is too quiet	Gap 0.57 against 0.47; deficit by frame loudness	Tab. 14, Tab. 8
AudioSR is dispersed but biased by band	Per-band levels; rank tails	Fig. 1, Fig. 3
Every model pulls each utterance’s level toward the corpus mean	Slope of deficit against the truth’s high-band share	Tab. 16
The deficit is set mostly by the loss domain, and closed by an energy score	Eight paired arms of one network	Tab. 6, Tab. 8

2 Data and protocol

Corpus. CSTR VCTK 0.92, microphone 1, resampled to 48 kHz with `scipy.signal.resample_poly`, peak-normalised to 0.95. Our training set contains every p-speaker with `id < 350` except p236–p238: 39,639 utterances, 37.3 h. The low-rate input for training is `resample_poly` decimation by 4 (48→12 kHz). Speakers with `id ≥ 350` (LISA’s validation split) and `s5` are excluded from our training, so every speaker of the released models’ test split is unseen by our arms.

Test sets. All three use the evaluation input of FLowHigh and NU-Wave 2: an order-8 Chebyshev I low-pass (0.05 dB ripple, cut-off 6 kHz) applied forward-backward (`sosfiltfilt`), then `resample_poly` to 12 kHz. The input is written as 32-bit float, so no model sees quantisation the others do not. Truth is trimmed to a multiple of 3840 samples, so every model’s hop divides it.

set	speakers	selection	utterances	seconds	draws per utterance
wide	p360–p364, p374, p376, s5	30 per speaker, evenly spaced	240	731	1
core	same	4 per speaker, evenly spaced	32	88	8
ourtest	p236, p237, p238	10–11 per speaker	30	165	16 (ours), 1 (released)

Table 1: Test sets. Levels, LSD, SNR and perceptual scores are reported on `wide`; every probabilistic metric on `core`.

Gain convention. Each output is scaled so that its energy below 5.5 kHz matches the truth’s. A model that copies the low band exactly gets gain 1, and its high band is then judged at the level the model placed it relative to its own low band.

Who saw which speakers. Table 2. The `ourtest` speakers are held out from our training but are in the training sets of FLowHigh, NU-Wave 2 and AP-BWE, so `ourtest` comparisons with those models are not fair to our arms. AudioSR’s training data includes VCTK with no stated speaker split, so it may have seen every test speaker.

model	wide/core speakers	our test speakers	source
ours (all eight arms)	unseen	unseen	our split
FLowHigh, NU-Wave 2, AP-BWE	unseen	seen	NU-Wave 2 split (100 training speakers)
AudioSR (speech)	possibly seen	possibly seen	multi-domain data including VCTK

Table 2: Training-set overlap with the test speakers.

3 Model and training

Network. LISA: a 1-D convolutional encoder (channels 16, 32, 64, 32; kernels 7, 3, 3, 1; ReLU) turns the 12 kHz waveform into one latent per input sample. A 5-layer ReLU MLP of width 144 then decodes the amplitude at a continuous time coordinate from $[t, z_{i-1}, z_i, z_{i+1}]$.

- **LISAS** concatenates 8 i.i.d. $\mathcal{N}(0, 1)$ channels to the encoder input, at the input rate: 87,777 parameters.
- **LISASD** also appends 4 $\mathcal{N}(0, 1)$ channels per 48 kHz output sample to the decoder input: 88,353 parameters. This follows DISCO Nets and EnScale in putting noise deeper in the network.

A draw is one forward pass with fresh noise; at noise temperature $\tau = 0$ the network is deterministic. The deterministic arms use the same class and are trained with the noise held at zero.

Objective. With y the target and y_1, y_2 two draws for the same input, every stochastic arm minimises the two-draw energy score

$$\mathcal{L} = \frac{1}{2} [d(y, y_1) + d(y, y_2)] - \frac{1}{2} d(y_1, y_2), \quad d = d_{\text{wave}} + \lambda d_{\text{spec}}. \quad (1)$$

- $d_{\text{wave}}(a, b)$ is the mean absolute difference of the waveforms.
- d_{spec} averages over three Hann STFTs, $(n_{\text{fft}}, \text{hop}) \in \{(2048, 512), (1024, 256), (512, 128)\}$.
- On each STFT it takes the mean of $|\log(|A| + 10^{-7}) - \log(|B| + 10^{-7})|$ ($\text{es_marg}, \text{es_dec}$).
- In the ERB arms ($\text{es_erb}, \text{es_dec_erb}$), it takes half of that plus half the same L1 on 32 ERB-band log energies, $\frac{1}{2} \log(W|S|^2 + 10^{-8})$, where W is a 32-band ERB filterbank from 50 Hz.

The deterministic arms minimise $d(y, \hat{y})$; `det_paper` is LISA’s released configuration (L1, $\lambda = 0$). The estimator is unbiased (§5.3).

Arms. Table 6. All eight share one batch stream: the sampler is seeded by a fixed label, so every arm sees identical segments in identical order, and differences between arms are paired.

Optimisation.

- Adam, learning rate 10^{-3} , halved at 20, 40, 50, 60, 70 and 80% of training (LISA’s schedule scaled to our step count), so the final rate is $\text{lr}_0/64$.
- Gradient-norm clipping at 10^{-3} , per arm.
- Batch 64 one-second segments; 104,950 steps, which is 50 epochs of the 37.3 h corpus.
- bf16 autocast on the network, all distances and STFTs in fp32, TF32, fused Adam. The decoder’s anchor jitter is kept from LISA.
- One arm per NVIDIA RTX PRO 6000 (Blackwell) GPU: 105 ms per step, about 3.1 h per arm.
- One training seed per arm.

Training curves are in Fig. 6; full histories and logs are in `results/training/`.

4 Metric definitions

All metrics come from one scorer, `code/sota/score.py`, applied to every model’s saved outputs. The high band (HB) is 6–24 kHz. Per-draw quantities are averaged over draws, then over utterances.

Deficit

The evaluation STFT is Hann, 1024/256. Take third-octave bands from 6 kHz to 24 kHz (centres 6.7, 8.5, 10.7, 13.4, 16.9 and 21.3 kHz). In each band compute $10 \log_{10}(E_{\text{out}}/E_{\text{true}})$ over all frames, then average over bands. 0 is right; negative is too quiet.

- *deficit bb* is the energy-weighted ratio over the whole band.
- *loud / mid / quiet* restrict to the truth’s frames in the top 25%, middle 50% and bottom 25% of frame magnitude.

LSD

Hann 2048/512, $\log_{10} |X|^2$, RMS over frequency, mean over frames. This is the basis of NU-Wave 2’s `for_test.py`. LSD-HF and LSD-LF split at 6 kHz.

SNR

$10 \log_{10}(\|y\|^2/\|y - \hat{y}\|^2)$ per whole utterance, averaged in dB over utterances.

- *Ceiling*: the truth with its HB removed (a brick wall at 6 kHz).
- *vs ceiling*: the model’s SNR minus that ceiling.

Coherent fraction, κ

Computed per HB third-octave band on the STFT, then averaged over bands: $c = \text{Re}\langle Y, W \rangle / \langle Y, Y \rangle$ (the predictable fraction) and $\kappa = \text{Re}\langle Y, W \rangle / \langle W, W \rangle$ (1 = informative energy, 0 = energy unrelated to the truth).

CRPS

The features are HB log-magnitudes $\log(|X| + 10^{-8})$ (natural log; one unit is 8.69 dB) on the 1024/256 STFT, bins $k \geq 128$. For each bin, with M draws $x_1, \dots, x_M$ and truth y :

$$\text{CRPS}_{\text{fair}} = \frac{1}{M} \sum_j |x_j - y| - \frac{1}{2M(M-1)} \sum_{j \neq k} |x_j - x_k|.$$

This is averaged over bins, frames and utterances. The tables’ “CRPS” column for an ensemble is this fair estimator. The appendix also reports the all-pairs estimator, which divides by M^2 ; its expected value exceeds the infinite-ensemble CRPS by $\mathbb{E}|X - X'|/(2M)$ [4]. A one-output model scores its mean absolute error, which is the CRPS of a point forecast.

Sliced CRPS

The all-pairs CRPS on 32 fixed random unit projections of each frame’s HB log-magnitude vector. It is sensitive to dependence across bins.

PIT tails

The rank of the truth among the M draws in each HB bin, pooled over bins and utterances. *PIT low* and *PIT high* are the shares where the truth is below all draws and above all draws. For a calibrated ensemble each is $1/(M+1) = 0.111$ at $M = 8$.

Spread

The pooled ratio $\sum_u s_u^2(1 + 1/M)/\sum_u \|y_u - \bar{x}_u\|^2$, computed on the HB waveform, where s_u^2 is the unbiased ensemble variance and $\bar{x}_u$ the ensemble mean. It is 1 for a calibrated ensemble and below 1 when the ensemble is under-dispersed.

Gap

$10 \log_{10}(\text{mean}_i \|y - x_i\|^2 / \|y - \bar{x}\|^2)$ on the HB waveform. For a calibrated ensemble it is $10 \log_{10}(2/(1 + 1/M)) = 2.50$ dB at $M = 8$ (§5.2).

corr err

$\|C_{\text{true}} - C_{\text{draw}}\|_F$, where C is the correlation matrix of 16 HB frequency groups of log-magnitude, pooled over frames and utterances.

Slope

The least-squares slope of the per-utterance deficit (dB) against the truth’s HB energy share (dB), on `wide` ($n = 240$). A slope of 0 means the model tracks each utterance’s brightness; -1 means it emits a fixed level.

Perceptual

ViSQOL v3 in audio mode at 48 kHz (and NSIM), and in speech mode at 16 kHz; wide-band PESQ. All on draw 0.

5 Derivations

5.1 The high-band error and the SNR ceiling

Let the truth’s HB have energy E , the output’s HB energy pE , and $c = \text{Re}\langle Y_H, W_H \rangle / E$. With a perfect low band,

$$\|y - \hat{y}\|^2 = \|Y_H - W_H\|^2 = E + pE - 2cE = E(1 + p - 2c). \quad (2)$$

An empty band ($p = c = 0$) costs exactly E , so its SNR is the ceiling $C \approx -10 \log_{10}(\text{HB share})$. A model beats C only if $c > p/2$. An incoherent band at the right level ($p = 1, c \approx 0$) costs $10 \log_{10} 2 = 3.0$ dB. This matches the measured -2.8 dB of FLowHigh, -3.1 dB of AP-BWE and -2.8 dB of our sampler against the ceiling on wide (Tab. 12).

Conversely, a reported SNR S bounds the HB level: $p \leq 10^{(C-S)/10} - 1$, since any low-band error only lowers S ; with nonzero coherence the bound is $p \leq 10^{(C-S)/10} - 1 + 2c$. As a consistency check, take NU-Wave 2’s own Table 1, where the “Input” row is its ceiling: at 12 kHz, $C - S = 0.5$ dB, and with its measured $c = 0.009$ this gives $p \leq 0.14$, i.e. at most -8.5 dB. We measure a broadband deficit of -8.9 dB. The identity reads a published SNR as a statement about high-band level, and our harness agrees with NU-Wave 2’s own number. The ceiling measured on our 240 utterances is in Table 11, and reported SNRs against it in Fig. 2.

5.2 Averaging draws, and the calibration target

For draws $x_1, \dots, x_M$ with HB components that are mutually incoherent and of equal energy, $\mathbb{E}\|\bar{x}_H\|^2 = \mathbb{E}\|x_H\|^2/M$: the waveform mean loses $10 \log_{10} M$ dB. A regressor trained on squared error estimates the $M \rightarrow \infty$ mean, which is the argument for why point training yields a quiet band.

If y and the x_i are exchangeable given the input (a calibrated sampler), $\mathbb{E}\|y - x_i\|^2 = 2\sigma^2$ and $\mathbb{E}\|y - \bar{x}\|^2 = \sigma^2(1 + 1/M)$. This gives the gap target $10 \log_{10}(2/(1 + 1/M))$ and the spread normalisation above.

The log-domain mean behaves differently. If a bin’s power is exponentially distributed, as for a circular Gaussian bin, then $\mathbb{E} \log P = \log \mathbb{E} P - \gamma$. The per-bin mean of log-magnitudes therefore sits $10\gamma/\ln 10 = 2.51$ dB below the draws’ energy as $M \rightarrow \infty$, where γ is the Euler–Mascheroni constant.

Scorer check on synthetic data. Build a set whose HB draws are fresh noise of the true envelope and level, so the truth is exchangeable with the draws by construction.

- `score.py` returns gap 2.50 dB (target 2.50), spread 0.999 (target 1), and PIT tails 0.109 and 0.125 (target 0.111).
- `mean_of_M.py` returns waveform-mean deficits of -3.01 , -6.02 and -9.03 dB at $M = 2, 4, 8$ (predicted -3.01 , -6.02 , -9.03), and a log-mean deficit of -2.63 dB at $M = 8$ (limit -2.51).

`code/sota/mean_of_M.py` computes the mean-of- M table on real draws, and `code/sota/run_all.sh` runs it.

5.3 The two-draw energy score: unbiased and proper

For any d , $\mathbb{E}[\frac{1}{2}(d(y, Y_1) + d(y, Y_2)) - \frac{1}{2}d(Y_1, Y_2)] = \mathbb{E}d(y, Y) - \frac{1}{2}\mathbb{E}d(Y, Y')$, the energy score, with no finite-ensemble bias [7, 8]. Its propriety depends on d .

Here d is an L_1 sum over waveform samples and over STFT and ERB features. The energy score of a sum of coordinatewise absolute differences is the sum of per-coordinate CRPS terms. It is therefore proper, and strictly proper for the marginal distribution of each coordinate, but not for their joint distribution: the L_1 metric on $\mathbb{R}^n$ is of negative type but not of strong negative type.

The multi-scale STFT and ERB terms add transformed coordinates, which keeps the score proper [1], but dependence between bins is constrained only through those transforms. This is why the sliced CRPS and the band-correlation error are reported next to the per-bin CRPS.

5.4 CRPS of a point forecast

For a point forecast m , CRPS equals $\mathbb{E}|Y - m|$. For any m , $\frac{1}{2}\mathbb{E}|Y - Y'| \leq \mathbb{E}|Y - m|$, so a calibrated sampler has lower expected CRPS than every point forecast. The comparison between samplers and deterministic models on CRPS (Table 6) is therefore expected by construction. The informative comparisons are among ensembles, and between ensembles and the calibration targets.

5.5 One Euler step of FLOWHigh returns the conditional mean

FLOWHigh’s path (`independent_cfm_adaptive`) trains on $x_t = tx_1 + (1-t)x_0 + \sigma_t\epsilon$ with $\sigma_t = 1 - (1 - \sigma_{\min})t$ and target $u = (x_1 - x_0) - (1 - \sigma_{\min})\epsilon$. Here x_0 is the input’s mel spectrogram, x_1 the target’s, and $\sigma_{\min} = 10^{-4}$.

In training, the network at $t = 0$ sees $x = x_0 + \epsilon$ together with x_0 (as conditioning), so $\epsilon = x - x_0$ is identified. The optimal field is therefore $v^*(x, 0) = \mathbb{E}[x_1 | x_0] - x_0 - (1 - \sigma_{\min})(x - x_0)$. Starting instead from $x = x_0 + s\epsilon$, one Euler step returns

$$x + v^*(x, 0) = \mathbb{E}[x_1 | x_0] + \sigma_{\min} s \epsilon, \quad (3)$$

the conditional mean for any start-noise scale s , then a deterministic vocoder. This follows from the rectified-flow identity $v(x, t) = \mathbb{E}[X_1 - X_0 | X_t = x]$ [17]. Restoring the trained prior ($s = 1$) cannot restore the spread at one step (§6.1).

6 The released models

Each model runs through its own inference code on the same input (`code/sota/run_models.py`), with the seed set per draw.

model	release used	sampling	notes
FLOWHigh	official code (MIT), FLOWHigh_indep_adaptive_400k + BigVGAN 48 kHz	1 Euler step (their setting)	deterministic as shipped (§6.1)
NU-Wave 2	official code (BSD), official checkpoint	8 DDIM steps, their schedule	loaded without pytorch-lightning; weights unchanged
AP-BWE	official code (MIT), 12→48 kHz checkpoint	one pass	not level-invariant; fed at the stored level
AudioSR	official package, speech checkpoint	50 DDIM steps, guidance 3.5 (CLI defaults)	its low band also varies between draws

Table 3: Released models as run. AP-BWE was also scored on its own paper’s sinc input (“own sinc input” rows).

model	LSD reported	LSD ours	ViSQOL reported	ViSQOL ours	reported in
FLOWHigh	0.75	0.761	3.61	3.61	FLOWHigh, Tab. I
AP-BWE	0.78	0.780	3.46	3.48	AP-BWE, Tab. V
NU-Wave 2	0.94	0.989	2.75	2.57	AP-BWE, Tab. V (re-run)

Table 4: Does the harness reproduce each model’s own numbers? wide, 240 utterances. Reported values are from the papers’ 12→48 kHz VCTK tables (NU-Wave 2’s as re-run by AP-BWE; its own paper reports SNR, 21.6 dB, against our 21.3 dB).

6.1 FLOWHigh

The released `inference.py` calls `sample(..., std_2 = 1.)` without `std_1`. Inside `sample()`, if `std_1` is `None` or `std_2` is `None`: `std_1 = 1.0; std_2 = self.sigma`. So the start point is $x_0 + 10^{-4}\epsilon$, while training starts from $x_0 + \epsilon$ ($\sigma_0 = 1$). The README recommends `-time_step 1 -sigma 0.0001`. The explicit `std_2 = 1.` suggests the override is unintended. `code/patches/flowhigh_inference_restore_prior.diff` is the one-line change that keeps the trained prior.

Measured on `core`:

- As shipped, the HB spread ratio is 2×10^{-9} .
- With the prior restored it is 0.023, and the truth falls below or above all eight draws in 41% and 46% of HB bins (Fig. 3).

This is what §5.5 predicts: the override is real, but one-step sampling is what forces the mean. In Table 7, FLOWHigh’s deficit, LSD and ViSQOL are as shipped, and its spread and CRPS are from the prior-restored run, the most favourable reading for it.

6.2 Cost

model	params	RTF	device	sampling
FLowHigh	49.4 M	1.237	cpu	1 Euler step (their setting)
AP-BWE	29.8 M	0.071	cpu	1 pass
NU-Wave 2	1.7 M	4.985	cpu	8 DDIM steps
AudioSR (speech)	672.0 M	4.611	mps	50 DDIM steps

Table 5: Parameters, and real-time factor (wall seconds per second of audio) on an Apple M4, each model alone, warm, median of three. Our 88 k network runs at about 0.017 on the same CPU.

7 Results

7.1 Full versions of the paper’s tables

arm	one change	deficit (dB)	step	spread	CRPS
det_paper	LISA recipe (L1, $\lambda=0$)	−21.78	–	–	2.853
det	+ log-magnitude term, $\lambda=0.01$	−6.18	15.6 [det_paper]	–	0.889
es_marg	point → draw (energy score)	−1.05	5.1 [det]	0.526	0.537
es_dec	+ decoder noise	−1.12	−0.1 [es_marg]	0.499	0.535
es_erb	+ ERB-band term	−1.08	0.0 [es_marg]	0.442	0.512
es_erb, $\lambda=0.1$	$\lambda \times 10$	−0.63	0.5 [es_erb]	0.479	0.515
es_erb, $\lambda=0.001$	$\lambda/10$	−5.73	−4.6 [es_erb]	0.225	0.683
es_dec_erb, $\lambda=0.1$	decoder noise + ERB, $\lambda=0.1$	−0.70	–	0.464	0.514

Table 6: All eight arms of one network (the paper’s Table 2 shows six). Deficit on *wide* (240 utterances, one draw); step = change of deficit against the arm in brackets. Spread and CRPS on *core* (32 utterances, 8 draws); a deterministic arm’s CRPS is its MAE.

model	params	deficit (dB)	LSD	ViSQOL	spread	CRPS	slope
FLowHigh	49.4 M	−1.60	0.761	3.61	0.023	0.682	−0.07
AP-BWE	29.8 M	−1.38	0.780	3.48	–	0.777	−0.12
NU-Wave 2	1.7 M	−7.96	0.989	2.57	0.094	0.583	−0.32
AudioSR (speech)	672.0 M	−0.60	1.561	2.41	0.765	1.243	−0.34
det	88 k	−6.18	0.864	3.13	–	0.889	−0.22
es_dec_erb, $\lambda=0.1$	88 k	−0.70	0.895	2.83	0.464	0.514	−0.23

Table 7: Released models against two of our arms (the paper’s Table 1, with spread and slope). Deficit, LSD and ViSQOL (audio mode) are on *wide*; spread and CRPS on *core*; slope on *wide* ($n = 240$ for every model). FLowHigh’s spread and CRPS are with its prior restored.

7.2 Levels

condition	deficit	deficit bb	loud	mid	quiet	LSD	LSD-HF	LSD-LF
FLowHigh	-1.60	-2.33	-1.71	-1.34	-1.03	0.761	0.869	0.212
FLowHigh, prior restored	-2.38	-3.09	-2.45	-2.05	-1.28	0.761	0.869	0.211
AP-BWE	-1.38	-1.23	-1.74	-0.90	0.18	0.780	0.894	0.179
AP-BWE, own sinc input	-1.51	-1.41	-1.84	-1.16	-0.16	0.781	0.893	0.212
NU-Wave 2	-7.96	-8.93	-8.96	-5.17	4.72	0.989	1.117	0.351
AudioSR (speech)	-0.60	2.17	-1.03	0.28	-5.20	1.561	1.778	0.470
det_paper	-21.78	-15.66	-20.89	-23.50	-21.30	2.319	2.674	0.210
det	-6.18	-7.38	-6.74	-3.87	-2.26	0.864	0.987	0.237
es_marg	-1.05	-1.34	-1.90	1.53	-2.20	0.929	1.036	0.432
es_dec	-1.12	-1.45	-2.26	1.87	-1.74	0.919	1.032	0.399
es_erb	-1.08	-1.64	-1.86	1.60	-1.17	0.902	1.012	0.398
es_erb, $\lambda=0.1$	-0.63	-1.31	-1.63	1.98	-1.20	0.894	1.012	0.326
es_erb, $\lambda=0.001$	-5.73	-4.07	-6.25	-3.41	-3.83	1.025	1.109	0.668
es_dec_erb, $\lambda=0.1$	-0.70	-1.44	-1.53	1.87	-1.58	0.895	1.012	0.336

Table 8: wide: deficits (dB) and LSD for every condition. Loud, mid and quiet are the truth’s frames by loudness. NU-Wave 2’s band is -9.0 dB on loud frames and +4.7 dB on quiet ones: it stays on between words.

condition	deficit	deficit bb	loud	mid	quiet	LSD	LSD-HF	LSD-LF
FLowHigh	-1.18	-1.95	-1.08	-1.41	-0.89	0.760	0.868	0.215
FLowHigh, prior restored	-1.86	-2.61	-1.74	-2.05	-1.15	0.760	0.868	0.215
AP-BWE	-1.11	-1.07	-1.23	-0.59	0.16	0.775	0.888	0.180
AP-BWE, own sinc input	-1.20	-1.24	-1.27	-0.87	-0.17	0.776	0.887	0.212
NU-Wave 2	-7.85	-8.73	-8.61	-5.46	4.24	0.982	1.109	0.346
AudioSR (speech)	-0.99	2.24	-1.20	-0.07	-4.87	1.552	1.761	0.519
det_paper	-21.31	-15.23	-20.48	-23.17	-22.06	2.350	2.709	0.208
det	-5.82	-7.04	-6.28	-3.63	-2.37	0.863	0.985	0.236
es_marg	-0.72	-1.08	-1.53	1.88	-1.88	0.924	1.032	0.429
es_dec	-0.98	-1.34	-1.93	1.86	-1.68	0.914	1.027	0.396
es_erb	-1.08	-1.70	-1.80	1.68	-1.32	0.895	1.005	0.393
es_erb, $\lambda=0.1$	-0.31	-1.07	-1.17	2.07	-1.35	0.891	1.009	0.324
es_erb, $\lambda=0.001$	-5.67	-4.13	-6.13	-3.44	-4.37	1.029	1.112	0.672
es_dec_erb, $\lambda=0.1$	-0.29	-1.10	-1.02	2.03	-1.73	0.891	1.007	0.335

Table 9: core, the utterances on which calibration is measured.

condition	deficit	deficit bb	loud	mid	quiet	LSD	LSD-HF	LSD-LF
FLowHigh	−1.00	−1.00	−1.11	−0.26	0.40	0.740	0.844	0.224
AP-BWE	−3.12	−2.95	−3.26	−2.35	0.06	0.779	0.893	0.182
det_paper	−25.16	−20.88	−25.05	−24.34	−18.63	2.167	2.498	0.206
det	−10.16	−11.56	−11.17	−6.76	−0.75	0.885	1.012	0.237
es_marg	−4.53	−5.05	−5.90	−0.79	−0.23	0.958	1.066	0.443
es_dec	−4.77	−5.31	−6.34	−0.77	0.19	0.937	1.049	0.410
es_erb	−4.73	−5.43	−6.01	−0.92	0.82	0.926	1.038	0.405
es_erb, $\lambda=0.1$	−4.13	−4.97	−5.53	−0.41	0.63	0.914	1.034	0.331
es_erb, $\lambda=0.001$	−8.87	−7.49	−9.90	−5.44	−1.18	1.029	1.118	0.629
es_dec_erb, $\lambda=0.1$	−4.26	−5.19	−5.66	−0.51	0.34	0.914	1.033	0.340

Table 10: ourtest (p236–p238). These speakers are in the training sets of FLOWHigh and AP-BWE (Table 2).

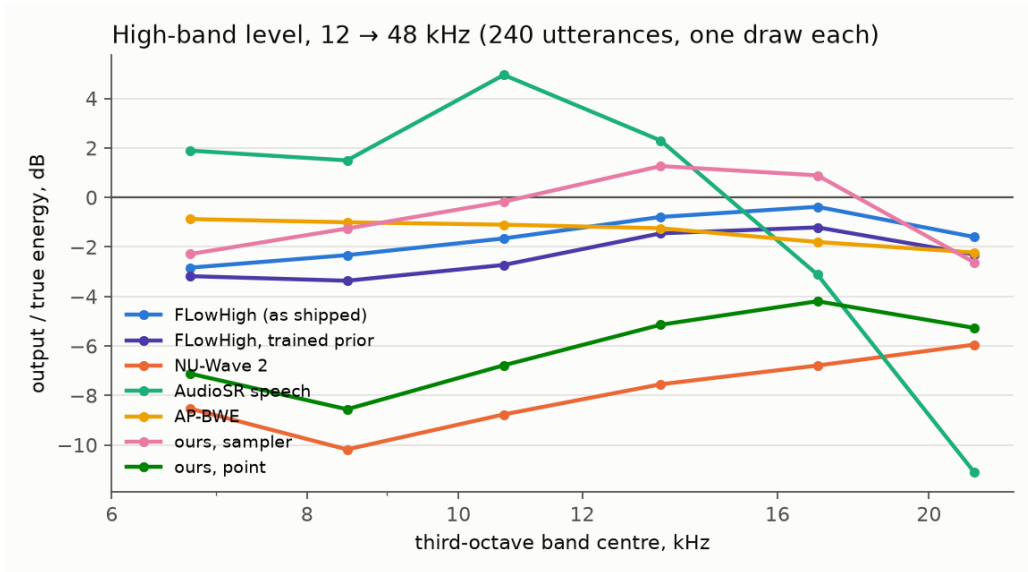


Figure 1: High-band level per third-octave band on wide. AudioSR’s small mean deficit hides +5 dB at 11 kHz and −11 dB at 21 kHz.

7.3 SNR, the ceiling, and coherence

input rate	empty-band SNR ceiling (dB)
8 kHz	19.00
12 kHz	22.02
16 kHz	24.67
24 kHz	30.00

Table 11: Empty-band SNR ceiling on wide (true low band, nothing above the input’s Nyquist frequency), per input rate.

condition	SNR (dB)	vs ceiling	coherent frac.	κ
FLowHigh	19.18	−2.84	0.0014	0.002
FLowHigh, prior restored	19.45	−2.57	0.0015	0.003
AP-BWE	18.95	−3.07	0.0164	0.020
AP-BWE, own sinc input	19.05	−2.96	0.0217	0.027
NU-Wave 2	21.26	−0.76	0.0094	0.065
AudioSR (speech)	15.44	−6.58	−0.0005	0.000
det_paper	21.87	−0.15	0.0105	0.145
det	21.19	−0.83	0.0143	0.071
es_marg	18.83	−3.18	0.0126	0.014
es_dec	18.89	−3.13	0.0125	0.015
es_erb	18.96	−3.06	0.0094	0.016
es_erb, $\lambda=0.1$	19.17	−2.85	0.0145	0.022
es_erb, $\lambda=0.001$	18.59	−3.43	0.0125	0.021
es_dec_erb, $\lambda=0.1$	19.26	−2.76	0.0151	0.025

Table 12: wide. Every model sits below the ceiling. The coherent fraction is below 0.02 for every model on the Chebyshev input; AP-BWE on its own sinc input reaches 0.022.

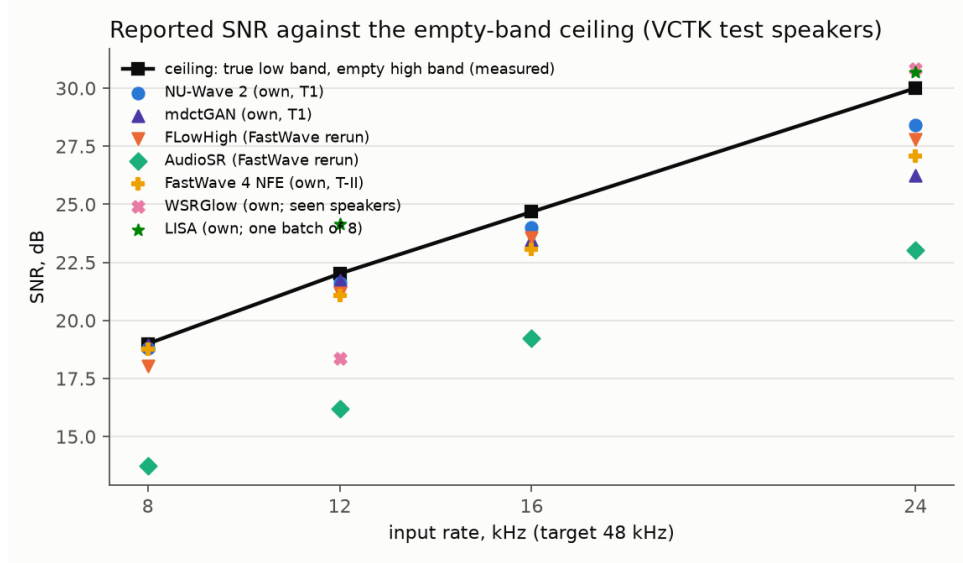


Figure 2: Reported SNRs from the literature against the ceiling measured on the same test speakers. Markers above the line come from protocols that differ from the ceiling’s, as labelled (speakers seen in training, or one batch of eight clips).

7.4 Calibration and proper scores

condition	gap HB (dB)	spread	PIT low	PIT high	CRPS	CRPS fair	sliced	corr err
FLowHigh	0.00	0.000	0.487	0.513	0.756	0.756	0.784	0.478
FLowHigh, prior restored	0.08	0.023	0.408	0.460	0.692	0.682	0.705	0.500
NU-Wave 2	0.57	0.094	0.132	0.217	0.638	0.583	0.614	0.863
AudioSR (speech)	2.62	0.765	0.110	0.520	1.287	1.243	1.204	0.782
es_marg	2.21	0.526	0.098	0.241	0.591	0.537	0.576	0.374
es_dec	2.13	0.499	0.098	0.237	0.589	0.535	0.572	0.383
es_erb	2.03	0.442	0.111	0.211	0.564	0.512	0.555	0.289
es_erb, $\lambda=0.1$	2.07	0.479	0.108	0.213	0.569	0.515	0.559	0.415
es_erb, $\lambda=0.001$	1.16	0.225	0.097	0.421	0.726	0.683	0.677	0.476
es_dec_erb, $\lambda=0.1$	2.07	0.464	0.107	0.212	0.567	0.514	0.560	0.438

Table 13: `core`, $M = 8$. Ideals: gap 2.50 dB, spread 1, PIT low = PIT high = 0.111, corr err 0. “CRPS” is the all-pairs estimator; “CRPS fair” is the one used in the paper.

The two waveform-domain measures do not always agree. Our sampler’s gap (2.07 dB) is within 17% of the calibrated 2.50 dB, while its pooled spread ratio (0.46) says it is under-dispersed. Both are dominated by output level when the band is incoherent with the truth (§5.1), so neither is read here as a pure measure of diversity.

condition	HB level (dB)	gap	gap if indep.	spread	spread if indep.	PIT low	PIT high
FLowHigh	−1.9	0.00	1.81	0.000	0.665	0.487	0.513
FLowHigh, prior restored	−2.6	0.08	1.61	0.023	0.577	0.408	0.460
NU-Wave 2	−8.7	0.57	0.47	0.094	0.148	0.132	0.217
AudioSR (speech)	2.2	2.62	3.45	0.765	1.557	0.110	0.520
es_marg	−1.1	2.21	2.10	0.526	0.800	0.098	0.241
es_dec	−1.3	2.13	2.01	0.499	0.756	0.098	0.237
es_erb	−1.7	2.03	1.89	0.442	0.702	0.111	0.211
es_erb, $\lambda=0.1$	−1.1	2.07	2.10	0.479	0.801	0.108	0.213
es_erb, $\lambda=0.001$	−4.1	1.16	1.21	0.225	0.415	0.097	0.421
es_dec_erb, $\lambda=0.1$	−1.1	2.07	2.09	0.464	0.795	0.107	0.212

Table 14: Level against dispersion, `core`, $M = 8$. When $c \approx 0$, draws that are mutually independent and carry p times the true HB energy give gap $10 \log_{10} \frac{1+p}{1+p/M}$ and spread $\frac{p(1+1/M)}{1+p/M}$ (from §5.1), with p from the measured HB level. A model whose gap matches the independent value is diverse at its own level, whatever its spread: NU-Wave 2 (0.57 against 0.47) and our arms are; FLowHigh (0.08 against 1.61 with its prior restored) is not.

condition (one output)	CRPS = MAE	sliced	corr err
AP-BWE	0.777	0.807	0.230
AP-BWE, own sinc input	0.775	0.807	0.230
det_paper	2.853	2.334	1.355
det	0.889	0.894	0.507

Table 15: One-output models on `core`: their CRPS is their MAE.

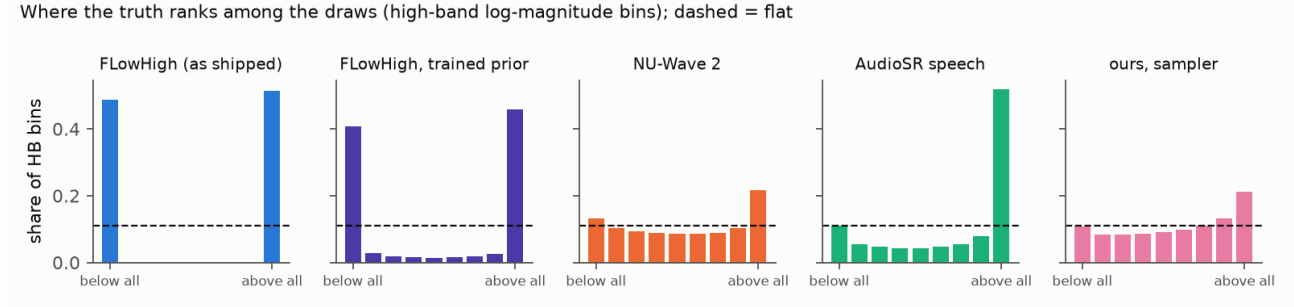


Figure 3: Rank of the truth among eight draws, HB log-magnitude bins, `core`. Dashed: flat. FLOWHigh collapses to a point. AudioSR puts the truth above every draw in 52% of bins, a level bias. Our sampler’s excess is one-sided (top tail 0.21, bottom 0.11): a residual level bias rather than too narrow an ensemble.

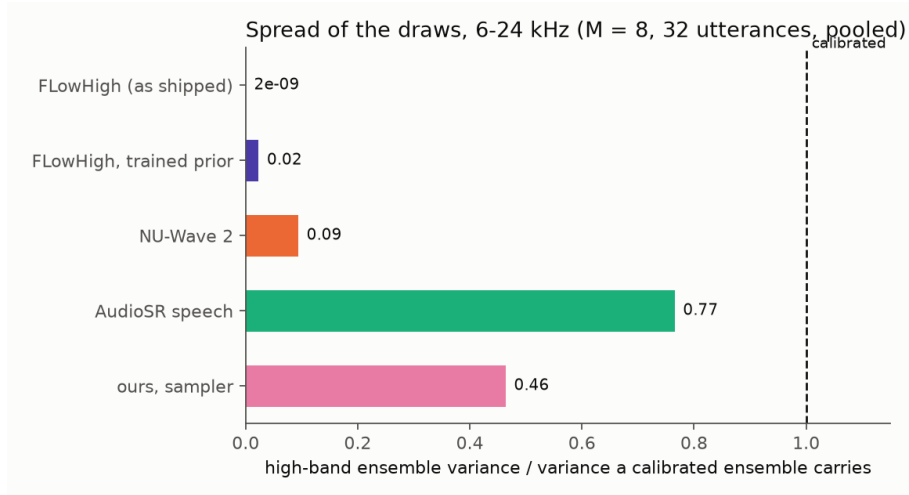


Figure 4: HB spread ratio, `core`. 1 = calibrated.

7.5 Level against brightness

model	slope (dB/dB)	r	n
FLowHigh	−0.075	−0.16	240
AP-BWE	−0.119	−0.26	240
NU-Wave 2	−0.315	−0.48	240
AudioSR (speech)	−0.338	−0.47	240
det_paper	−0.346	−0.43	240
det	−0.223	−0.36	240
es_marg	−0.239	−0.39	240
es_dec	−0.261	−0.41	240
es_erb	−0.287	−0.48	240
es_erb, $\lambda=0.1$	−0.251	−0.38	240
es_erb, $\lambda=0.001$	−0.196	−0.31	240
es_dec_erb, $\lambda=0.1$	−0.235	−0.36	240

Table 16: Slope of the per-utterance deficit against the truth’s HB share, `wide`. The two models that read a whole frame or more (FLowHigh, AP-BWE) track the utterance; every objective we trained sits between −0.20 and −0.35, whatever its mean level.

7.6 Perceptual scores

condition	ViSQOL audio	NSIM audio	ViSQOL speech	PESQ wb
FLowHigh	3.61	0.901	4.25	4.46
FLowHigh, prior restored	3.62	0.902	4.25	4.47
AP-BWE	3.48	0.881	4.05	4.36
AP-BWE, own sinc input	3.49	0.885	4.12	4.36
NU-Wave 2	2.57	0.771	3.07	3.95
AudioSR (speech)	2.41	0.780	3.73	3.89
det_paper	1.85	0.745	3.94	4.45
det	3.13	0.853	4.12	4.35
es_marg	2.77	0.812	3.81	4.05
es_dec	2.79	0.817	3.90	4.07
es_erb	2.82	0.821	3.96	4.09
es_erb, $\lambda=0.1$	2.86	0.828	4.02	4.04
es_erb, $\lambda=0.001$	2.63	0.771	3.20	3.73
es_dec_erb, $\lambda=0.1$	2.83	0.823	4.01	4.12

Table 17: wide, draw 0. ViSQOL audio mode spans 1.57 (empty band) to 4.73 (true band) on this task. Speech-mode ViSQOL and PESQ stop at 8 kHz and see only 6–8 kHz of the missing band.

Quality against size and speed, 12 -> 48 kHz (32 utterances, 8 unseen speakers). Grey: non-dominated models.

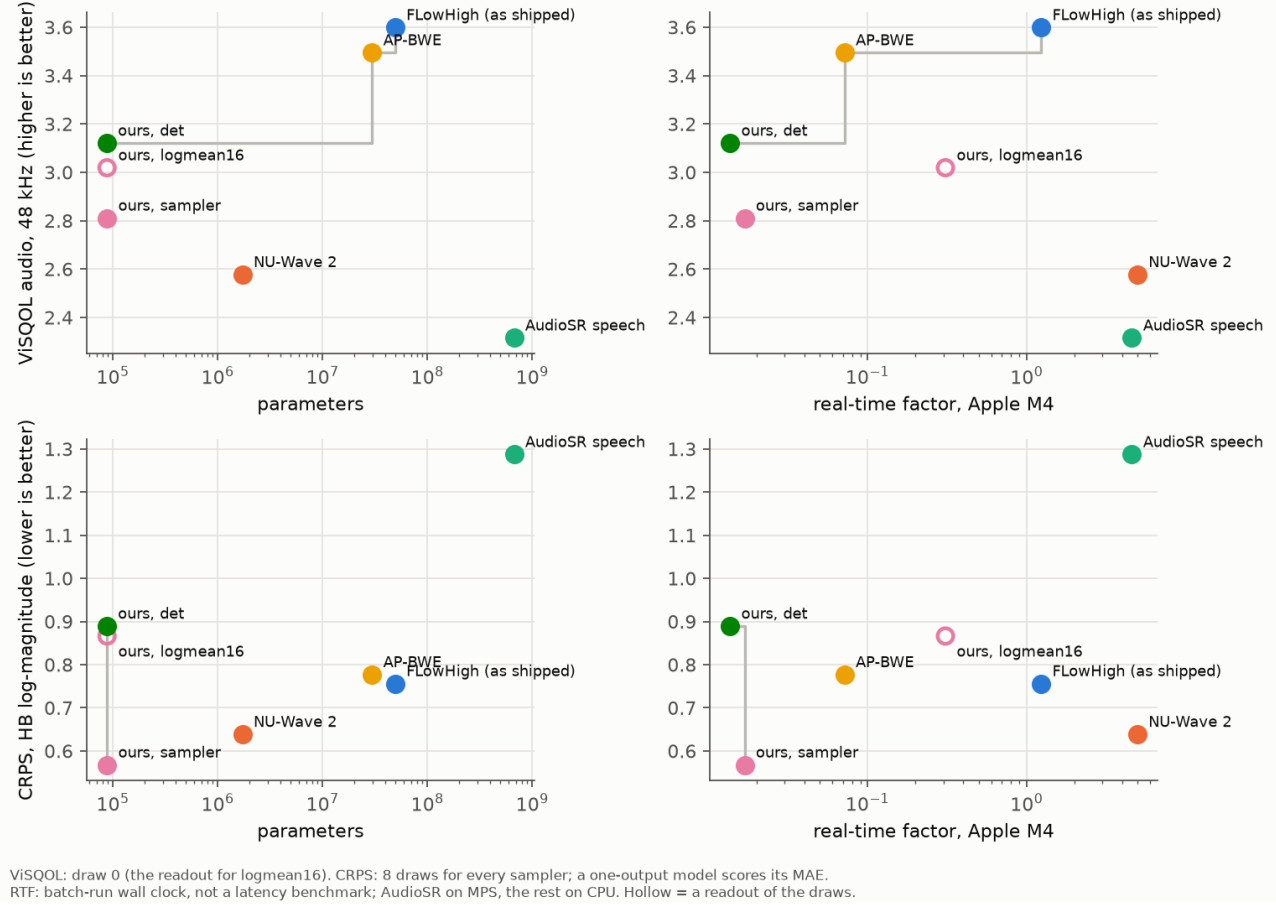


Figure 5: Quality against size and speed, core. This figure uses the all-pairs CRPS and FLOWHigh as shipped; Table 7 uses the fair estimator and FLOWHigh with its prior restored.

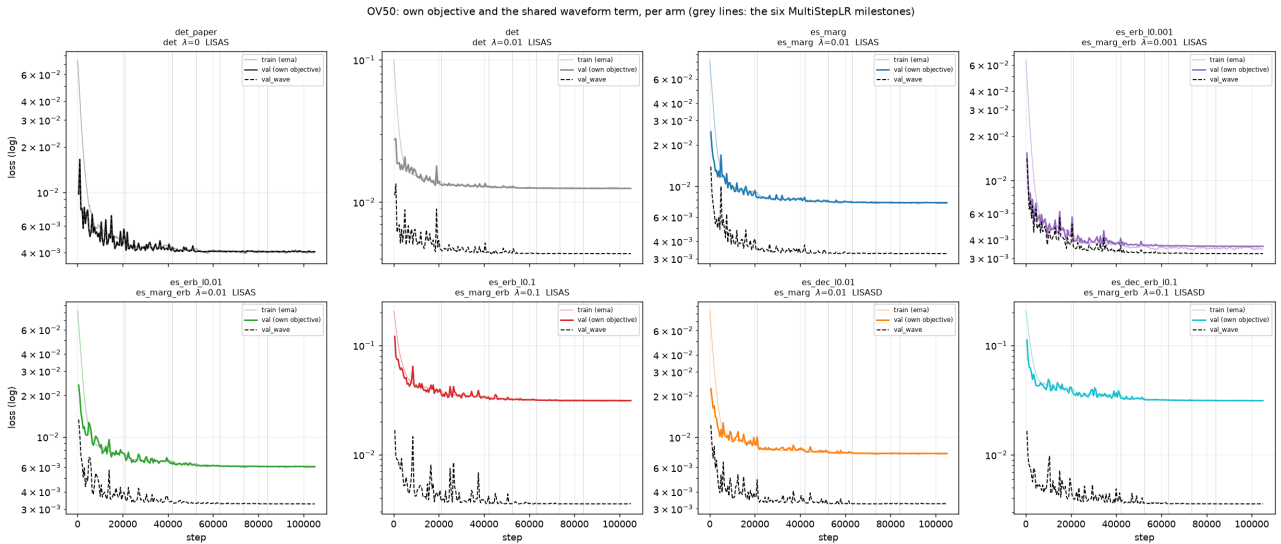


Figure 6: Training and validation curves of the eight arms (one run each, identical batches).

8 Limitations

- **No listening test.** ViSQOL and LSD favour FLowHigh and AP-BWE; CRPS favours the samplers. Which ordering listeners share is not settled here. A proper score ranks predictive distributions, not the perceptual quality of one draw.
- **Small calibration set, one seed.** 32 utterances $\times$ 8 draws; one training seed per arm. Ranks and bins are correlated within an utterance, so binomial error bars would be too narrow. Utterance-level bootstrap intervals need the per-utterance score files, which `score.py` writes but which are not included here.
- **Scored features.** CRPS is scored per bin on HB log-magnitude. It is proper for each marginal, not for the joint (§5.3), and the logarithm weights quiet bins heavily. Sliced CRPS and corr err cover part of the joint.
- **Different spaces.** Spread and gap are measured on the HB waveform; CRPS and PIT on log-magnitude.
- **Input-filter mismatch.** Our arms were trained on `resample_poly` input and tested on the Chebyshev input. On p236–p238 this moves the deficit by 0.04–0.07 dB.
- **Training-data overlap.** `ourtest` is seen by three released models, and AudioSR’s data may include every test speaker (Table 2).
- **One test corpus.** VCTK read speech only, 12→48 kHz only for the calibration results.

9 Related work: proper scoring and this argument

- **BWE evaluation.** Evaluation sections of 119 BWE and audio super-resolution papers (2017 to September 2026) were searched. None evaluates with a strictly proper scoring rule or a calibration diagnostic, and none reports a held-out likelihood, including models trained by likelihood (WaveNet BWE, WSRGlow, UDM+, CVAE BWE).
 - Diffusion restoration work suggests the spread of several draws as an uncertainty measure [16].
 - Inference-time search measures spread across AudioSR draws but treats it as noise to select over [13].
 - BABE and CQT-Diff add set-level Fréchet distances [19].
- **Other fields.** The protocol used here (CRPS, rank histograms, spread-skill) is standard in stochastic downscaling of weather and climate fields, which is super-resolution by another name [14, 10, 18]. A general inverse-problem version was proposed in 2026 [2].
- **Earlier statements of the argument.** The steps before it are established:
 - BWE is ill-posed [12, 3];
 - regression returns a muffled mean [23, 15, 6];
 - SNR rewards an empty band [22, 9].

Speech synthesis also found that samples beat the mean only when the model is accurate [11], and classical BWE biased the band quieter on purpose, because over-estimation is judged more harshly [20].

Our contribution is the measurement applied to released BWE systems, and what it finds there.

10 Reproducing

1. **Environment.** `code/README.md` lists the three Python environments. It also gives two CPU tests (31 and 18 checks), which pass on this snapshot.
2. **Train.** `code/fast/train_arm.py`, one arm per GPU, about 3 h each.
3. **Evaluate.** `code/sota/run_all.sh`: inputs, released models, our arms, the scorer, the mean-of- M table and the figures.
4. **Tables.** `results/make_tables.py` rebuilds every table in this appendix and checks each number printed in the paper against `results/results.json` (`results/number_check.md`: all 68 match).

References

- [1] S. Allen, D. Ginsbourger, J. Ziegel. Evaluating forecasts for high-impact events using transformed kernel scores. *SIAM/ASA J. Uncertainty Quantification* 11, 2023.
- [2] M. H. Baatrup et al. Pointwise metrics mislead: an evaluation protocol for multimodal inverse problems. arXiv:2605.22891, 2026.
- [3] J. Bruna, P. Sprechmann, Y. LeCun. Super-resolution with deep convolutional sufficient statistics. ICLR 2016.
- [4] C. A. T. Ferro, D. S. Richardson, A. P. Weigel. On the effect of ensemble size on the discrete and continuous ranked probability scores. *Meteorological Applications* 15, 2008.
- [5] D. Freirich, T. Michaeli, R. Meir. A theory of the distortion-perception tradeoff in Wasserstein space. NeurIPS 2021.
- [6] S.-W. Fu et al. Rethinking training targets, architectures and data quality for universal speech enhancement. arXiv:2603.02641, 2026.
- [7] T. Gneiting, A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *JASA* 102, 2007.
- [8] A. Gritsenko et al. A spectral energy distance for parallel speech synthesis. NeurIPS 2020.
- [9] S. Han, J. Lee. NU-Wave 2: A general neural audio upsampling model for various sampling rates. Interspeech 2022.
- [10] L. Harris et al. A generative deep learning approach to stochastic downscaling of precipitation forecasts. *JAMES* 14, 2022.
- [11] G. E. Henter, S. King, T. Merritt, G. Degottex. Analysing shortcomings of statistical parametric speech synthesis. arXiv:1807.10941, 2018.
- [12] P. Jax, P. Vary. An upper bound on the quality of artificial bandwidth extension of narrowband speech signals. ICASSP 2002.
- [13] Y. Jin et al. Inference-time scaling for diffusion-based audio super-resolution. AAAI 2026.
- [14] J. Leinonen, D. Nerini, A. Berne. Stochastic super-resolution for downscaling time-evolving atmospheric fields with a generative adversarial network. *IEEE TGRS* 59, 2021.
- [15] J.-M. Lemerrier, J. Richter, S. Welker, T. Gerkmann. StoRM: A diffusion-based stochastic regeneration model for speech enhancement and dereverberation. *IEEE/ACM TASLP* 31, 2023.
- [16] J.-M. Lemerrier et al. Diffusion models for audio restoration: a review. *IEEE Signal Processing Magazine* 41(6), 2024.
- [17] X. Liu, C. Gong, Q. Liu. Flow straight and fast: learning to generate and transfer data with rectified flow. arXiv:2209.03003, 2022.
- [18] M. Mardani et al. Residual corrective diffusion modeling for km-scale atmospheric downscaling. *Communications Earth & Environment* 6, 2025.
- [19] E. Moliner, F. Elvander, V. Välimäki. Blind audio bandwidth extension: a diffusion-based zero-shot approach. *IEEE/ACM TASLP* 32, 2024.
- [20] M. Nilsson, W. B. Kleijn. Avoiding over-estimation in bandwidth extension of telephony speech. ICASSP 2001.
- [21] G. Ohayon, T. Adrai, M. Elad, T. Michaeli. Reasons for the superiority of stochastic estimators over deterministic ones: robustness, consistency and perceptual quality. ICML 2023.
- [22] J. Su, Y. Wang, A. Finkelstein, Z. Jin. Bandwidth extension is all you need. ICASSP 2021.
- [23] H. Zen, K. Tokuda, A. W. Black. Statistical parametric speech synthesis. *Speech Communication* 51, 2009.